DEX-Net:
A Generative Model of Uniswap v3 Order Flow

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Abstract

Liquidity providers (LPs) in concentrated-liquidity automated market makers (AMMs) need simulators of how swaps, mints and burns arrive and interact. We present DEX-Net, a neural marked temporal point process for Uniswap v3. It learns which event will come next, when the next event will come, and how large the next event will be. An exact replica of the pool then computes the resulting state which is acted upon recursively by the model. We are interested in optimizing simulation performance across regimes of dearth and ample historical data. To this end, we train on 29.4 million events from ten Ethereum pools and ask three questions: (1) does computing the pool mechanics exactly, instead of learning them, improve simulation; (2) what improves model generalization to an unseen pool; and (3) what is the best model for pools with substantial training data? We find that (1) exact mechanics forecast 512 events ahead 11–20% better than learned price and liquidity changes on non-stablecoin pools; (2) giving the model the liquidity near the current price, reduces small-error compounding; and (3) training on all pools is best: it beats training on the pool alone, most clearly for pools with a short history, and fine-tuning on the pool of interest is not reliably better. We compare DEX-Net to multiple baseline models and show by most accounts its superior performance. For a hypothetical LP position, DEX-Net forecasts fee income better than history baselines at most horizons. We release the code, model weights and dataset.

1 Introduction

A liquidity provider (LP) in a Uniswap v3 pool earns fees from swaps that trade through its price range and loses value to arbitrageurs who trade against stale pool prices. The two effects depend on swap frequency and size relative to active liquidity. Price-only simulators, such as geometric Brownian motion, GARCH-type models (Bollerslev, 1986) and learned return generators (Wiese et al., 2020), do not explicitly model pool liquidity or swap arrivals and sizes. Agent-based simulators capture liquidity dynamics through specified agent behaviors and calibrated parameters (Angeris et al., 2021; Di Nosse and Lillo, 2026).

We take a third route. An AMM is a deterministic machine: given its state and an event, the next state follows exactly from the protocol's rules. What is uncertain is which, when, and how large of an event arrives. We learn only this stochastic part from historical data and compute what is deterministic exactly. A simulated path is a sequence of sampled events, each applied to an exact replica of the pool.

We ask three questions. First, does computing the pool mechanics exactly, instead of learning them, improve simulation? Second, what does a model need to simulate a pool it was not trained on, for which there is no history to learn from? Third, for a pool that does have its own history, which model forecasts it best? We also ask a question about method: is next-event likelihood, the usual criterion for training and selecting such models, a reliable guide to how well they simulate?

In brief, the answers are as follows. Exact mechanics make the simulation more accurate on the eight training pools, except on stablecoin pools: exact models forecast 512 events ahead 11–20% better than a version that learns each event's effect on price and liquidity, and their simulated liquidity stays realistic. To simulate an unseen pool, the model needs the current liquidity profile, which the exact replica provides; without it, small errors in sampled swap size compound through the exact mechanics, and on a held-out stablecoin pool the 512-event energy score is 127 without the profile and 7.6 with it. For a pool with its own history, the best choice is the model trained jointly on all pools: it beats a model trained on that pool alone, most clearly when the pool's history is short, and fine-tuning it on the pool was not reliably better across seven pools. Next-event likelihood is a poor guide: the model with the lowest next-event NLL has the worst rollouts.

Contributions.

1. A reconciled multi-pool Uniswap v3 event dataset: ten Ethereum pools from May 2021-September 2026, with active liquidity reproduced exactly at every swap (Section 3).
2. DEX-Net, an event model trained jointly across pools, with exact, mechanics-consistent sampling and liquidity profile conditioning (Section 4), with ablations measuring what exact mechanics, the liquidity profile and joint training each contribute (Section 6.7)
3. A multi-seed evaluation of next-event likelihood against point-process baselines, rollouts against history-based forecasters, liquidity behavior, and the fee income and P&L of a hypothetical LP position, including a confirmation in AMM simulation that next-event likelihood can be a poor guide to rollout quality, and the mechanism behind it (Section 6).

2 Background and related work

Uniswap v3. In Uniswap v3 (Adams et al., 2021), LPs deposit liquidity L over a price range $[p_a, p_b)$ expressed in discrete ticks. Within a tick interval, the pool behaves as a constant-product market with active liquidity equal to the sum of all positions covering the current price; a swap moves the price along this curve, crossing ticks and changing active liquidity as it goes. Mints and burns add and remove liquidity over ranges. Fees, a fixed fraction of swap input, accrue to in-range liquidity. Uniswap governance turned on a protocol share of v3 fees on Ethereum mainnet in December 2025, first for selected pools, and in February 2026 proposed extending it to the remaining v3 pools (Uniswap Governance, 2025, 2026). The share is one quarter of fees on the 0.01% and 0.05% tiers and one sixth on the 0.3% tier. Activation dates differ by pool, so we report all LP results gross of this share.

LP economics. Loss-versus-rebalancing (LVR; Milionis et al., 2022) measures what an LP loses to arbitrageurs relative to a rebalancing strategy trading at external prices; with fees, the LP's excess return over the strategy is fees minus LVR. Just-in-time (JIT) liquidity, where an LP mints immediately before a large swap and burns immediately after, is a documented strategy in Uniswap v3 (Wan and Adams, 2022). Impermanent loss in concentrated ranges is analyzed by Loesch et al. (2021).

Temporal point processes. Event sequences with marks are classically modeled as Hawkes processes (Hawkes, 1971); neural variants replace parametric intensities with recurrent or attention-based encoders (Mei and Eisner, 2017; Zuo et al., 2020), or model inter-event times directly without

Table 1: Pools, event counts and chronological splits (model rows: every event after a pool's first 64, which lack a full history). Held-out pools contribute no training rows.

Pool	Fee	Role	Events	Swaps	Mints	Burns	Dates	Train	Val	Test
USDC/WETH	0.05%	fit	12,064,159	11,652,190	219,779	192,190	2021-05-05 – 2026-09-15	8,416,528	1,833,866	1,813,701
USDT/WETH	0.05%	fit	7,064,516	6,965,286	52,348	46,882	2021-05-05 – 2026-09-15	4,746,323	1,139,022	1,179,107
WBTC/WETH	0.05%	fit	2,390,754	2,330,001	33,297	27,456	2021-05-05 – 2026-09-15	1,298,702	538,585	553,403
DAI/WETH	0.05%	fit	1,985,785	1,976,002	5,231	4,552	2021-05-05 – 2026-09-15	1,160,931	376,088	448,702
USDC/USDT	0.01%	fit	1,960,944	1,908,221	27,508	25,215	2021-11-13 – 2026-09-15	1,089,657	482,479	388,744
USDT/WETH	0.3%	fit	1,539,097	1,333,943	115,486	89,668	2021-05-05 – 2026-09-15	796,809	311,711	430,513
USDC/WETH	0.3%	fit	1,177,154	984,031	110,214	82,909	2021-05-04 – 2026-09-15	795,457	183,125	198,508
DAI/WETH	0.3%	fit	414,307	392,844	12,347	9,116	2021-05-04 – 2026-09-15	275,445	69,620	69,178
WBTC/WETH	0.3%	held out	479,419	404,556	42,471	32,392	2021-05-04 – 2026-09-15	—	88,503	66,414
USDC/USDT	0.05%	held out	305,855	292,975	6,845	6,035	2021-05-04 – 2026-09-15	—	16,849	18,163
Total			29,381,990	28,240,049	625,526	516,415				

an intensity (Shchur et al., 2020). Our timing model falls in the last category with a discrete-time (block) parameterization.

Generative market simulators. Limit-order-book simulators range from agent-based platforms such as ABIDES (Byrd et al., 2020) to learned generators of order flow, such as neural Hawkes processes conditioned on the state of the book (Shi and Cartlidge, 2022), which are often checked against stylized facts of returns (Cont, 2001). For AMMs, Palit (2022) replays randomly shuffled historical Uniswap events through the AMM's mechanics to study price formation without trader rationality; like us it separates order flow from mechanics, but it does not learn a generator. To our knowledge, learned generators of AMM order flow that keep the AMM's mechanics exact have not been explored.

Likelihood and multi-step error. Models trained to predict one step ahead from real histories are known to degrade when rolled out on their own outputs: exposure bias in sequence models (Bengio et al., 2015; Ranzato et al., 2016), compounding error in learned time-series and dynamics models (Venkatraman et al., 2015), and a mismatch between the training objective and downstream use (Lambert et al., 2020). Likelihood and sample quality can also disagree for generative models in general (Theis et al., 2016). Neural point processes are nonetheless usually benchmarked on next-event likelihood. We confirm the gap in AMM simulation and identify the mechanism behind it (Section 6.5).

3 Data

Pools. We used ten Ethereum Uniswap v3 pools: USDC, USDT, DAI, and WBTC paired with WETH at the 0.05% and 0.3% fee tiers, and USDC/USDT at 0.01% and 0.05%. The data covers May 2021-September 2026, including every initialize, swap, mint, and burn (Table 1). This constitutes 29.4 million events comprising 28.24 million swaps, 625,526 mints and 516,415 burns.

Reconciliation. We replay every pool from its first event with exact integer arithmetic. A Fenwick-tree ledger of liquidity by tick reproduces the active liquidity recorded by the contract at every one of its 28.24 million swaps.

Liquidity profile. For every event we also record the liquidity profile just before it: the active liquidity at 17 log-price offsets between -4% and $+4\%$ of the current price, as $\log(1+L_{\text{offset}}/L_{\text{now}})/4$,

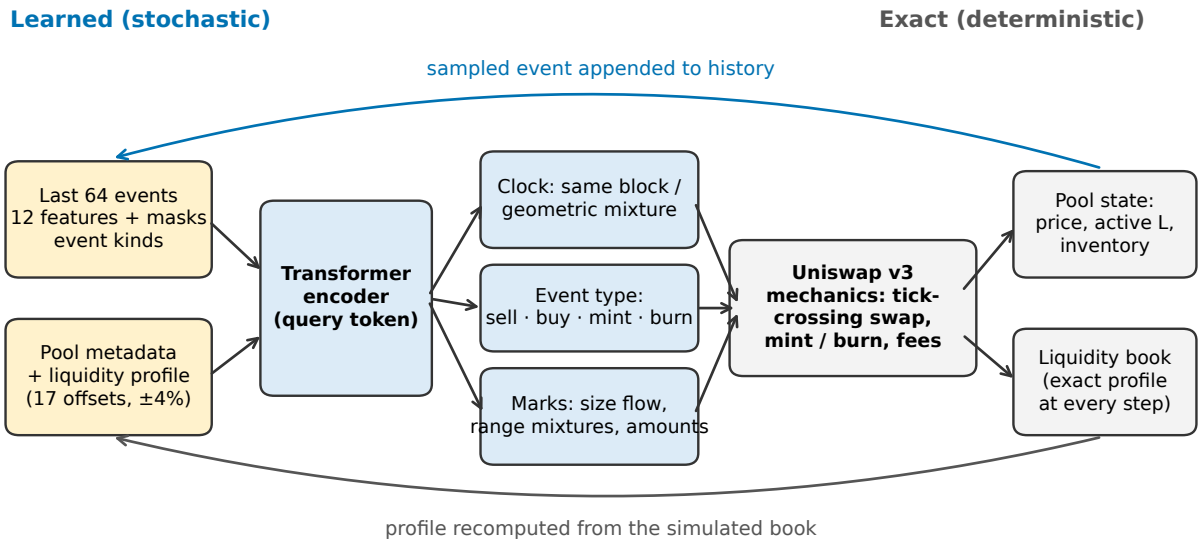


Figure 1: DEX-Net. A transformer encoder reads the last 64 events, pool metadata and the liquidity profile (omitted in the no-profile model), and parameterizes distributions over the next event's timing, type and marks. A sampled event is applied by exact Uniswap v3 mechanics. The state after a model step yields the next liquidity profile and the event joins the history.

scaled so that typical values fall roughly between 0 and 1. During training the profile is read from the reconciled ledger. During simulation it is computed from the simulated book after every event.

Splits. Splits are chronological: training until 2 February 2025, validation until 14 November 2025, and test until 15 September 2026. Eight pools are used for fitting while two are used for validation and test only: WBTC/WETH (0.3% fee) and USDC/USDT (0.05% fee).

4 Model

Figure 1 summarizes the model and the sampling loop.

Event space. Events may be a sell, buy, mint or burn. They may arrive after a gap of $d \in \{0, 1, 2, \dots\}$ blocks, where $d = 0$ means the same block as the previous event.

Clock. A logit gives the probability of another event in the same block. Otherwise the gap $d \geq 1$ follows a three-component geometric mixture, $P(d) = \sum_k \pi_k (1 - e^{-r_k})\, e^{-(d-1)r_k}$, with mixture weights π_k and rates r_k output by the network.

Type and marks. The event type is a four-way categorical conditioned on the gap. Marks are conditioned on the gap and type:

1. Swap size: a rational-quadratic spline flow (Durkan et al., 2019) on the log of the base amount relative to the virtual base reserve. The base token is the pool's economic base asset (e.g., WETH in USDC/WETH), and the virtual base reserve is $L/\sqrt{P}$, the amount of base the pool would hold at the current price if its active liquidity extended over all prices. A buy

removes base from the pool, so its size is truncated at the exact amount the pool's liquidity can supply. A sell adds base, which the pool can absorb up to its lowest initialized tick, so sells are not truncated; an oversized sell ends the path as a failure.

2. Mint range: discretized logistic mixtures over the lower tick (relative to the current tick on an asinh scale) and the width (log scale), on the pool's tick grid.

3. Mint amount: a spline flow on $\log(L/L_{\text{active}})$.

4. Burn: the same range model, normalized over the ranges that actually hold liquidity (the simulator tracks liquidity per range, summed over owners, alongside the tick-level ledger), followed by a full-burn indicator and a spline flow for partial fractions.

Encoder. A transformer reads the last 64 events (features, masks, and event-type embeddings) and a query token carrying pool metadata. DEX-Net projects the liquidity profile and adds it to the query token; it has width 128 and 4 layers (956,527 parameters). The no-profile model is identical but without the liquidity profile (954,351 parameters), and DEX-Net-small has width 64 and 2 layers (148,975 parameters).

Exact simulation. To simulate we sample an event, apply it to the replicated pool with the protocol's exact swap arithmetic (Adams et al., 2021) or exact liquidity updates, compute its features from the before and after states, and append it to the history. If a sampled event is impossible (a swap larger than the pool's liquidity can absorb, a mint above the protocol's per-tick liquidity limit, or a state with zero active liquidity), the path ends and is reported as a failure. Failures are rare: within 512 events at most 0.3% of the no-profile model's paths fail (held-out pools), and 0.02% or fewer of DEX-Net's. When scoring, failed paths are dropped from their ensemble and every anchor is kept.

Training. We minimize next-event negative log-likelihood (NLL), the sum of the clock, type and mark terms, with AdamW (learning rate 10^{-4}, weight decay 0.01, 1,000 update warm-up then cosine decay to 10%, and gradient clipping at 1). Each batch of 1,024 holds 256 events from each of 4 different pools, and every pool contributes the same number of examples. Because the pools have a different number of events, some were cycled through more than others (3 to 92 cycles). We calculate the validation score every 2,048 batches (one round) until we hit 3 cycles (epochs) through the largest pool. The best performing checkpoint (lowest seen-pool macro validation NLL) was kept.

5 Evaluation setup

We train five families with three seeds each: DEX-Net, the same model without the liquidity profile (the no-profile model), DEX-Net-small, and two history ablations of DEX-Net. Two further ablations, also with three seeds, are described in Section 6.7: the no-profile model with learned state transitions in place of the exact replica, and DEX-Net trained on a single pool.

Next-event likelihood. A macro averaged NLL is reported in nats per event on 32,768 fixed samples per pool from the test period.

Point-process baselines. For the clock and type terms we fit two baselines by maximum likelihood on the same event space, scored on the same rows. We fit an unconditional renewal model (constant same-block probability, a three component geometric mixture for gaps, constant type frequencies), and a four-type discrete time Hawkes process over the same last 64 events, with exponential kernels at two learned time scales, a closed-form per-block hazard for inter event block gaps, and type probabilities proportional to type intensities (the expected rate of each event type: a base rate plus a decaying boost from each of the last 64 events). Both are fitted per seen pool; for held-out pools, we fit one model on pooled seen-pool data.

Rollouts. Start points (anchors) are spaced uniformly in event index space within each split such that we can rollout up to 512 events without overlapping with another anchor. We take up to 64 anchors per pool and split (fewer where the period is too short for 64 non-overlapping windows), giving 610 test anchors (512 on seen pools, 98 on held-out pools) and 607 validation anchors (512 and 95); scores average over anchors, so each anchor counts equally. Each model seed draws 32 paths per anchor. At 32, 128 and 512 events we score four endpoints (at 512 events, traded volume contributes most to the score): price return (bp), log change in active liquidity, log elapsed blocks, and traded volume relative to the initial virtual reserve (log scale). Each endpoint is divided by a fixed scale (10 bp, 0.1, 1 and 0.001), set before evaluation to put the endpoints on a comparable footing, and the four are scored jointly with the energy score (Gneiting and Raftery, 2007) using the fair ensemble estimator (Ferro, 2014), which removes the penalty a finite ensemble pays for its size and so makes ensembles of different sizes (32 paths, 256 baseline windows) comparable. For single quantities (one endpoint, or an LP outcome in Section 6.6) we use the CRPS, the energy score's one-dimensional case, with the same correction. The correction assumes exchangeable ensemble members; history windows overlap and are serially dependent, so for the history baselines it is approximate. Model families are compared via paired bootstrap across anchors; these confidence intervals reflect variation across anchors, not across training seeds.

History-based forecasters. Two history-based forecasters use the pool's own past instead of a model. We create an ensemble of 256 past h-event windows from the same pool, all ending before the anchor and computed with the same endpoint definitions as simulated paths. 1) Recent windows start at evenly spaced points within the last 32h events; 2) expanding windows start at points drawn uniformly from the pool's entire prior history. Both are scored identically to the model. For held-out pools we also build a sibling-pool baseline, for the case of a new pool with no history of its own: the same recent and expanding windows, built from a seen pool with the same pair at another fee tier (WBTC/WETH 0.05% for WBTC/WETH 0.3%, and USDC/USDT 0.01% for USDC/USDT 0.05%), using only sibling events from blocks before the anchor.

6 Results

6.1 Next-event likelihood

Table 2 reports NLL by pool. The no-profile model has the lowest NLL on aggregate (seen 5.534, held-out 7.649): 0.008 nats per event lower than DEX-Net on seen pools (s.e. 0.004) and 0.018 lower on held-out pools (s.e. 0.011). Standard errors of differences combine the seed-to-seed spread of both models, $\sqrt{s_1^2/3 + s_2^2/3}$, treating each model's three seeds as independent. DEX-Net's NLL is 0.078 nats per event lower than DEX-Net-small's (64×2) on seen pools (s.e. 0.007) and 0.130 lower on held-out pools (s.e. 0.020).

Table 2: Next-event NLL (nats/event) on the test period, mean ± s.d. over 3 seeds; 32,768 fixed examples per pool.

Pool	DEX-Net	No profile	DEX-Net-small	DEX-Net, 8 events	DEX-Net, no history
USDC/WETH 0.05%	4.792 ± 0.005	4.775 ± 0.006	4.887 ± 0.009	4.974 ± 0.006	5.476 ± 0.030
USDT/WETH 0.05%	4.461 ± 0.003	4.454 ± 0.009	4.584 ± 0.005	4.640 ± 0.008	5.125 ± 0.007
WBTC/WETH 0.05%	4.998 ± 0.007	4.988 ± 0.009	5.099 ± 0.051	5.115 ± 0.012	6.047 ± 0.046
DAI/WETH 0.05%	4.715 ± 0.002	4.704 ± 0.004	4.823 ± 0.009	4.895 ± 0.005	5.506 ± 0.052
USDC/USDT 0.01%	5.902 ± 0.001	5.904 ± 0.003	5.952 ± 0.008	6.030 ± 0.006	6.531 ± 0.091
USDT/WETH 0.3%	7.260 ± 0.039	7.244 ± 0.027	7.310 ± 0.039	7.564 ± 0.029	9.204 ± 0.063
USDC/WETH 0.3%	6.603 ± 0.009	6.598 ± 0.012	6.623 ± 0.011	6.839 ± 0.009	8.083 ± 0.035
DAI/WETH 0.3%	5.606 ± 0.006	5.602 ± 0.005	5.684 ± 0.002	5.720 ± 0.003	6.696 ± 0.083
WBTC/WETH 0.3% (held out)	7.075 ± 0.027	7.070 ± 0.022	7.206 ± 0.031	7.274 ± 0.009	8.723 ± 0.041
USDC/USDT 0.05% (held out)	8.259 ± 0.041	8.228 ± 0.018	8.389 ± 0.031	8.631 ± 0.093	38.377 ± 19.626
Macro, seen pools	5.542 ± 0.005	5.534 ± 0.005	5.620 ± 0.010	5.722 ± 0.004	6.583 ± 0.024
Macro, held-out pools	7.667 ± 0.019	7.649 ± 0.003	7.797 ± 0.028	7.953 ± 0.046	23.550 ± 9.794

Table 3: Clock and event-type NLL (nats/event, macro over pools) on the same evaluation rows. Point-process baselines are fitted per seen pool; held-out pools use one fit on pooled seen-pool data. Neural rows are 3-seed means.

Model	Clock (seen)	Type (seen)	Sum (seen)	Clock (held out)	Type (held out)	Sum (held out)
Unconditional (renewal + type frequencies)	2.606	0.896	3.503	6.285	0.964	7.250
4-type Hawkes, 2 time scales	2.495	0.736	3.231	4.730	0.784	5.515
DEX-Net	2.422	0.669	3.091	4.456	0.668	5.124
No profile (neural)	2.421	0.667	3.088	4.451	0.667	5.119

History length matters. With its full 64-event history, DEX-Net's NLL is lower than with only the last 8 events (by 0.180 nats on seen pools, s.e. 0.004; 0.286 on held-out pools, s.e. 0.029) and lower than with no history (by 1.042 on seen pools, s.e. 0.014).

The no-history model fails on held-out pools (23.6 ± 9.8). The failure is concentrated in one component and one pool: event timing in USDC/USDT 0.05%, where the clock NLL reaches 12–47 nats instead of 1–2. Without history the model presumably cannot tell how active an unseen pool is. Its lowest validation NLL occurs after very few training rounds (validation rounds 2, 3 and 16 of 97 across seeds), after which it learns the training period's average activity at the expense of transfer to later periods.

Table 3 compares the clock and type terms with the point-process baselines. A Hawkes fit improves on the unconditional baseline (3.50 to 3.23 nats on seen pools, 7.25 to 5.52 on held-out pools), and DEX-Net improves on Hawkes by a further 0.14 and 0.39. On seen pools the gain splits evenly between timing (2.42 against 2.50) and event type (0.67 against 0.74); on held-out pools most of it is in timing (4.46 against 4.73). Marks have no point-process analogue; the no-history ablation is the corresponding comparison for them.

6.2 Rollouts

Figure 2 shows forecast cones at four randomly drawn test anchors, and Table 4 reports energy scores for all models and baselines.

Held-out pools. On held-out pools DEX-Net is much more accurate than the no-profile model. In the test period DEX-Net's 512-event energy is 16.63 against 60.30 for the no-profile model, 72% lower (paired difference −43.7, 95% CI −79.0 to −21.8); in validation it is 33.03 against 80.30 (−47.3, CI −69.1 to −28.8), and in the test period DEX-Net is significantly better at every horizon

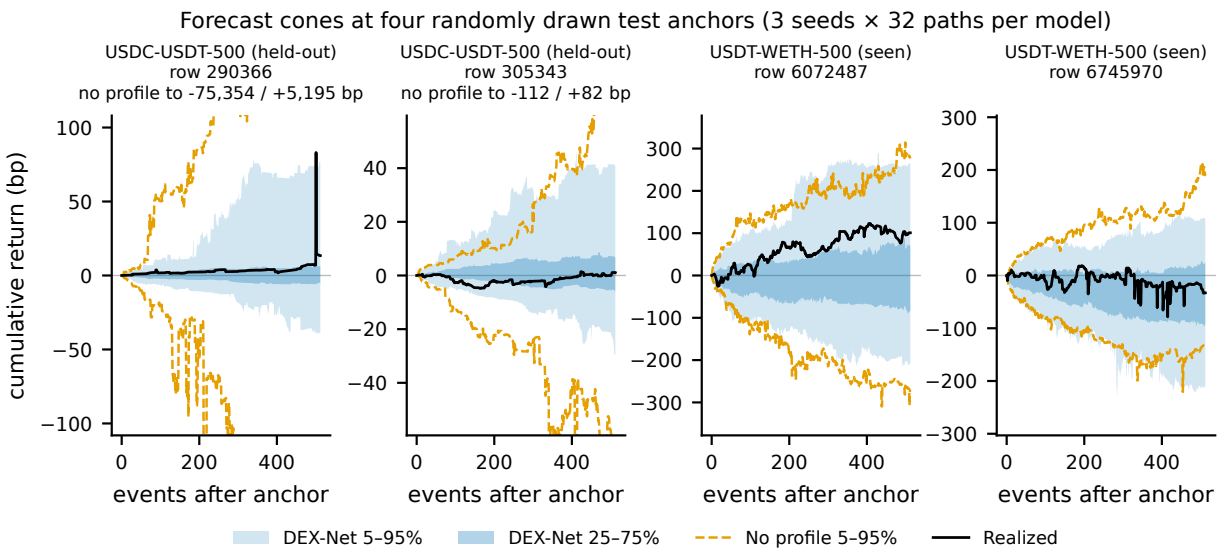


Figure 2: Forecast cones of cumulative return at four test anchors drawn at random (two held-out, two seen), from 96 paths per model (3 seeds × 32). Axes are scaled to DEX-Net and the realized path; where the no-profile model's 5–95% band leaves the axis its extent is printed in the panel title.

(Table 5). Confidence intervals that also resample training seeds are wider but keep this conclusion (test, 512 events: −43.7, CI −89.5 to −13.2). Most of the difference comes from one pool. On the held-out stablecoin pool (USDC/USDT 0.05%) DEX-Net's 512-event energy is 7.6 against 127.5 for the no-profile model in test, and 17.3 against 168.5 in validation. On WBTC/WETH 0.3% neither model is better in both periods: DEX-Net is 13.0% better in test (21.4 against 24.6; CI 7.5% to 18.1%), and the no-profile model is 7.5% better in validation (37.6 against 40.6; CI 2.6% to 12.6%). Section 6.7 shows that the stablecoin collapse comes from exact mechanics without the profile: a model that learns price and liquidity changes instead does not collapse. Against the strongest history baseline (recent windows), DEX-Net is 2.2% better in the test period (16.63 against 17.00; CI −9.6% to +14.6%), and the recent baseline is 4.7% better in validation (31.47 against 33.03; CI −9.7% to +21.9%). Neither difference is significant, and with 98 and 95 anchors from two pools the intervals include differences of 10–20% in either direction. By pool, DEX-Net is better than the recent baseline on WBTC/WETH (21.4 against 23.1 in test; 40.6 against 44.8 in validation), and the recent baseline is better on the stablecoin pool (5.6 against 7.6; 4.1 against 17.3), whose nearly static price the recent past predicts well. This baseline uses the held-out pool's own past, which a brand-new pool would not have. A pool's own recent history is better than its sibling's at every horizon in both periods, by 4–46% (significant except at 512 events on WBTC/WETH). Against the sibling's recent history, DEX-Net is better at 32 events (by 34% in test and 19% in validation, both significant) and at 128 events (by 17%, significant, and 9%, not significant), and the two are level at 512 events (DEX-Net better by 9.1%, CI −2.6% to +21.2%, and by 0.7%, CI −16.5% to +15.6%). On WBTC/WETH DEX-Net is better than the sibling's history at every horizon in both periods (512 events: by 10.8%, CI −1.2% to +23.4%, and 12.9%, CI 2.8% to 23.6%); on the stablecoin pool the sibling's history is better at 512 events in validation (5.6 against 17.3).

Table 4: Rollout energy score (lower is better) at 32/128/512 events on all anchors; a seed's forecast is the ensemble of its paths that reached the horizon (failed paths, at most 0.3%, are dropped); models are the mean of 3 seeds × 32 paths; best per row in bold.

Period	Pools	Events	Anchors	DEX-Net	No profile	DEX-Net-small	Recent	Expanding
Test	held out	32	98	**2.75**	3.43	3.01	3.15	3.61
Test	held out	128	98	6.38	12.46	**6.30**	6.48	7.23
Test	held out	512	98	**16.63**	60.30	16.64	17.00	17.76
Test	seen	32	512	**3.00**	**3.00**	3.01	3.57	4.10
Test	seen	128	512	7.88	**7.84**	8.23	8.00	10.82
Test	seen	512	512	26.29	26.61	31.44	**21.92**	32.81
Validation	held out	32	95	6.64	7.24	**6.24**	6.93	7.28
Validation	held out	128	95	11.81	15.71	11.24	**11.09**	12.97
Validation	held out	512	95	33.03	80.30	31.74	**31.47**	34.43
Validation	seen	32	512	**5.30**	5.33	5.35	6.22	6.51
Validation	seen	128	512	**12.11**	12.23	12.49	14.36	15.32
Validation	seen	512	512	**37.94**	39.27	41.38	44.12	46.05

Seen pools. On seen pools the two 128×4 models are within 3.4% of each other at every horizon (Table 5). The largest difference is at 512 events in validation, where DEX-Net is 3.4% better than the no-profile model (CI 0.6% to 6.6%); in test DEX-Net is 1.2% better (CI −1.3% to +3.7%). In validation DEX-Net is better than both history baselines at every horizon (512 events: 14.0% better than the recent baseline, 37.94 against 44.12). In the test period the recent baseline is 16.6% better at 512 events (21.92 against 26.29; CI 9.3% to 24.3%). Most of this difference is in the two DAI/WETH pools (energy differences of 14.5 and 19.2) and in the traded-volume endpoint (scaled CRPS 16.4 for the recent baseline against 21.3 for DEX-Net; for price return, 10.1 against 10.7). When pools are resampled as well as anchors, the interval includes zero (CI −1.0% to +37.2%). The comparison with the recent baseline reverses between the validation and test periods, which suggests that which forecaster is better depends on market conditions; the Market conditions analysis below relates this to how volatile each period was relative to the pools' history (Table 6).

Capacity. On held-out pools DEX-Net and DEX-Net-small are within 4% of each other at 512 events: DEX-Net is 0.1% better in test (CI −7.2% to +7.7%), and DEX-Net-small is 3.9% better in validation (CI −5.9% to +17.2%). On seen pools DEX-Net is better at long horizons: 16.4% better at 512 events in the test period (26.29 against 31.44; CI 13.0% to 20.1%).

Path reweighting. A post-hoc reweighting of simulated paths, fitted only on seen-pool validation rollouts (described in Appendix A), improves DEX-Net's 512-event energy on test anchors by 1.07 on seen pools (CI 0.74 to 1.45) and 1.14 on held-out pools (CI 0.46 to 2.10). Weighted ensembles are scored with a weighted analogue of the fair estimator that reduces to it for uniform weights (Appendix A), which puts these figures on the scale of Table 4; because the weights depend on the sampled paths, it carries no unbiasedness guarantee and serves as a practical adjustment. These comparisons use the anchors on which all of a seed's paths completed, so baseline differences can differ from Table 4 by a few hundredths. Reweighted, DEX-Net beats the recent baseline on held-out pools at 32 events (−0.48, CI −0.95 to −0.11) and is ahead, though not significantly, at 512 events (−1.59, CI −3.54 to +0.33); on seen pools the recent baseline is still better at 512 events (by 3.22, CI 1.46 to 5.14). Reweighting changes only weights, not paths, and we treat it as an optional

Table 5: DEX-Net minus the no-profile model: paired energy difference with 95% bootstrap CI over the Table 4 anchors; share of paths losing more than 63% of active liquidity; 5–95% return-spread ratio (simulated / realized).

Period	Pools	Events	Δ energy [95% CI]	Collapse real	Collapse DEX-Net	Collapse no profile	Spread DEX-Net	Spread no profile
test	seen	32	-0.00 [-0.04, +0.03]	1.0%	0.8%	0.8%	1.08×	1.06×
test	seen	128	+0.04 [-0.07, +0.15]	1.2%	1.6%	1.6%	1.25×	1.24×
test	seen	512	-0.32 [-0.98, +0.35]	1.6%	3.1%	3.9%	1.37×	1.46×
test	held out	32	-0.68 [-1.57, -0.11]	3.1%	4.0%	5.2%	0.95×	1.07×
test	held out	128	-6.08 [-14.20, -1.53]	2.0%	8.3%	13.0%	1.05×	1.29×
test	held out	512	-43.66 [-78.95, -21.84]	5.1%	15.2%	24.2%	1.01×	1.78×
val	seen	32	-0.03 [-0.08, +0.02]	2.0%	2.3%	2.3%	1.17×	1.16×
val	seen	128	-0.12 [-0.35, +0.07]	3.1%	3.2%	3.3%	1.02×	1.02×
val	seen	512	-1.34 [-2.58, -0.23]	3.5%	6.0%	6.8%	1.12×	1.21×
val	held out	32	-0.60 [-1.69, +0.03]	4.2%	4.6%	5.7%	0.93×	1.01×
val	held out	128	-3.90 [-7.90, -1.22]	4.2%	9.5%	14.8%	0.99×	1.13×
val	held out	512	-47.27 [-69.07, -28.81]	8.4%	19.9%	30.6%	0.80×	1.77×

Table 6: DEX-Net minus the recent baseline (512-event energy, mean over anchors with 95% bootstrap CI; negative favors DEX-Net), by thirds of recent volatility relative to the pool's long-run volatility.

Period	Pools	Calm third	Middle third	Volatile third
Test	seen	+7.17 [+3.40, +11.80]	+5.47 [+2.41, +8.51]	+0.48 [−1.96, +2.85]
Test	held out	+1.64 [−1.36, +4.88]	−1.58 [−5.35, +1.40]	−1.21 [−5.43, +2.83]
Validation	seen	−2.30 [−5.57, +0.41]	−4.80 [−8.83, −1.12]	−11.43 [−19.87, −5.65]
Validation	held out	+11.00 [+2.12, +24.66]	−0.56 [−8.29, +7.77]	−5.82 [−11.30, −0.57]

calibration layer rather than part of the model.

Market conditions. To see where DEX-Net beats the recent baseline, we split anchors into thirds by recent volatility relative to the pool's long-run volatility: the 5–95% width of 512-event returns in the recent windows divided by that in all earlier history, a quantity known at the anchor. DEX-Net's advantage grows with volatility in three of the four period and pool groups (Table 6); on held-out test pools no third differs significantly. On seen test pools the recent baseline is better by 7.2 in the calmest third, and the two are level in the most volatile third (difference 0.48, CI −1.96 to +2.85); in validation DEX-Net is better in every third, by 2.3 in the calmest (CI −0.4 to 5.6) and 11.4 in the most volatile. The recent baseline does well when the near future resembles the recent past, while DEX-Net, which models how order flow evolves from the current pool state, does better when volatility is high. This is not only shrinkage toward long-run behavior: against an equal blend of recent and long-run windows, which shrinks by construction, DEX-Net's advantage in the most volatile third remains in validation (seen pools −8.98, CI −15.70 to −4.29; held-out pools −6.36, CI −11.17 to −1.68), though not in the test period (seen pools −0.25, CI −2.04 to +1.45). The test period was calmer relative to each pool's history than the validation period (on seen pools the middle third spans ratios of 0.59–0.85 in test against 0.72–0.98 in validation), which is consistent with the reversal between periods. The analysis is post hoc and should be read as suggestive; because the ratio is known at the anchor, it could drive a forecast that relies more on DEX-Net when recent volatility is high.

Exploratory follow-up. Motivated by this pattern, we refit the reweighting with weights conditioned on the same volatility ratio. Because the choice was made after seeing the results above, this is exploratory. On the held-back part of the validation data the global reweighting scores better (reweighting objective, the mean over horizons of the weighted energy score relative to uniform weights, lower is better: 0.645 against 0.656; uniform weights 0.663). On test anchors it closes the 512-event energy gap on seen pools (+0.52 against the recent baseline, CI −0.88 to +1.97, from +3.22 with global weights), but its LP P&L forecasts become unstable across anchors (seen pools at 512 events: +8.97, CI −3.38 to +31.23). We therefore keep the global reweighting; a regime-aware calibration that stays stable remains open.

Forecast blend. Because DEX-Net and the recent baseline do well in different conditions, we also score a mixture of the two: a share α of DEX-Net's paths and $1 - \alpha$ of the baseline's windows. The mixture's energy score follows exactly from the within- and between-ensemble distances, so it is scored with the same fair estimator as Table 4. α is fitted per horizon on seen-pool validation anchors, either once (0.91, 0.87 and 0.82 at 32, 128 and 512 events) or per third of the volatility ratio above (at 512 events 0.67, 0.74 and 0.96 from the calmest to the most volatile third), and applied unchanged to test anchors, including held-out pools. With the per-third weights, the blend improves on DEX-Net at 512 events by 12.2% on seen pools (CI 9.3% to 15.8%) and 5.4% on held-out pools (CI 2.2% to 9.0%), by about 3% at 128 events, and not at 32. On seen pools the recent baseline is still better at 512 events, but by 5.1% (CI −0.6% to +10.6%) instead of the 16.6% by which it is better than DEX-Net alone. On held-out pools the blend is 7.4% better than the recent baseline (CI −1.8% to +18.0%). Like path reweighting, the blend is a post-hoc calibration layer rather than part of the model.

6.3 Liquidity behavior

Liquidity collapse and swap-size tails. The no-profile model's long-horizon failures on held-out pools come from paths in which simulated liquidity thins and then collapses. One-step swap sizes are close to calibrated (for example, the predicted share of swaps larger than 1% of the virtual reserve is 0.043% against 0.038% observed on seen test pools); the error is state-dependent. Because swap sizes are relative to the current virtual reserve, a swap's price impact within a range of constant liquidity does not depend on how much liquidity there is; the amplification comes from what lies beyond the current tick. When nearby ticks hold much less liquidity than the current one, a swap that crosses them moves the price much further than its size suggests, and each crossing thins the active liquidity for the next swap. The liquidity profile provides this information. On held-out pools, 24.2% of the no-profile model's paths lose more than 63% of their active liquidity within 512 events (a fall of more than 1 in log active liquidity, since $1 - 1/e \approx 63\%$), against 5.1% of real outcomes; DEX-Net reduces this to 15.2%. Almost all of this is on the stablecoin pool (no-profile 68%, DEX-Net 42%, real 15% in the test period); on WBTC/WETH collapse is rare for every model. Swaps that move the price by more than 100 bp make up 0.97% of the no-profile model's simulated swaps on held-out pools, 0.20% of DEX-Net's, and 0.03% of real swaps; for moves above 500 bp the shares are 0.48%, 0.066% and 0.018%. The liquidity profile removes most of the excess, but not all of it.

Just-in-time liquidity. We use the following definitions, applied identically to simulated paths and the real test-period event stream (Figure 3). A strict JIT pair is a mint, swap and burn that are consecutive, in one block, and share the same range and liquidity amount; a large mint adds more than ten times the active liquidity. The model learns the pattern: it generates strict JIT

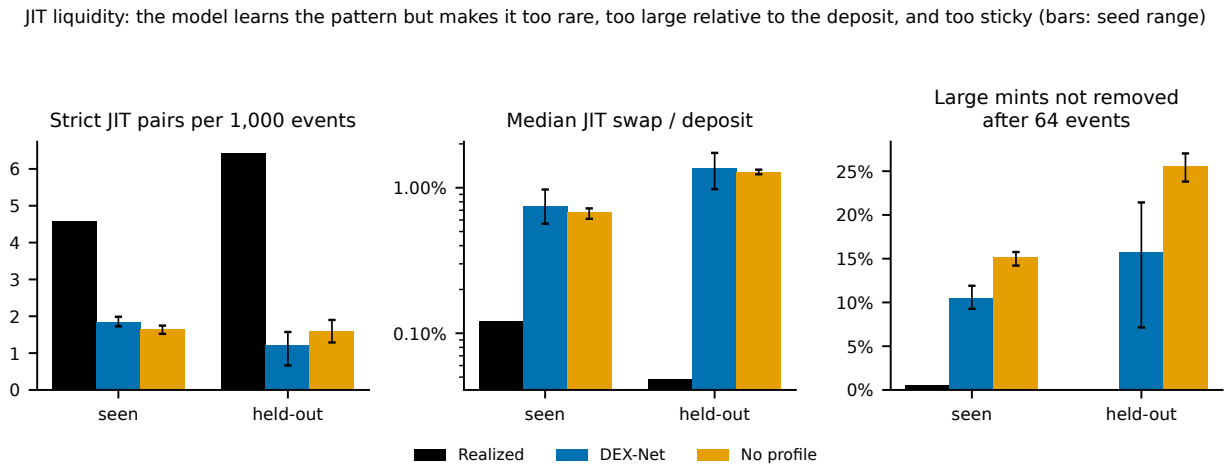


Figure 3: JIT liquidity, real against the final models, on seen and held-out pools: strict JIT pairs per 1,000 events, median ratio of the paired swap's notional to the minted deposit (log scale), and the share of large mints not removed within 64 events. Bars are seed means; whiskers show the range over three seeds. Simulated: 12 test anchors per pool, 32 paths of 96 events per seed. Real: the full test-period event stream.

pairs, and its share of large mints is close to real (seen pools: 7.5% of mints against 9.0%). But JIT is too rare (1.8 pairs per 1,000 events against 4.6 on seen pools; 1.2 against 6.4 on held-out pools), with deposits too small for the swap they accompany (median swap of 0.74% of the deposit against 0.12%), and too sticky (10.5% of large mints are not removed within 64 events, against 0.5%). The liquidity profile helps with the last of these (the no-profile model leaves 15.2% unremoved). Removal is measured only for large mints followed by at least 64 events (in simulation, mints within the first 32 of 96 events), and the same rule is applied to the real stream. Following each simulated large mint shows where the stickiness comes from. The next event is a swap in the same block about as often as in reality (97.7% against 98.3% on seen pools), and when the model burns the minted range it removes all of it in the same block, as real JIT liquidity is removed. The failure is the burn itself: 8.9% of simulated large mints on seen pools are never followed by a burn of their range within 16 events (real 1.1%), and in about half of these the model burns in the same block but a different position. Positions are identified by their range relative to the current price, so a range minted just before a price-moving swap has a different coordinate when it is burned, and among the hundreds of open positions in a WETH pool the minted one is hard to identify. On held-out pools 18.2% are never burned (real 0%), and the most common failure is letting time pass after the mint instead (54% of failed cases). The low JIT rate is partly a consequence, since a burn of a different range does not complete a strict pair. JIT timing also depends on pending swaps that providers see before they execute, which the executed history does not contain, so a model of this kind can match JIT's average rate but not its timing.

6.4 Stylized facts, calibration and skew

Figure 4 compares simulated and realized event sequences. The event mix is close; the same-block share is low on seen pools (0.18 against 0.23). On seen pools the width of the simulated return distribution is within about 12% of realized up to 32 events and widens to about 1.4 times realized at 512 events. On held-out test pools, DEX-Net's spread stays within roughly 0.9–1.3 times realized

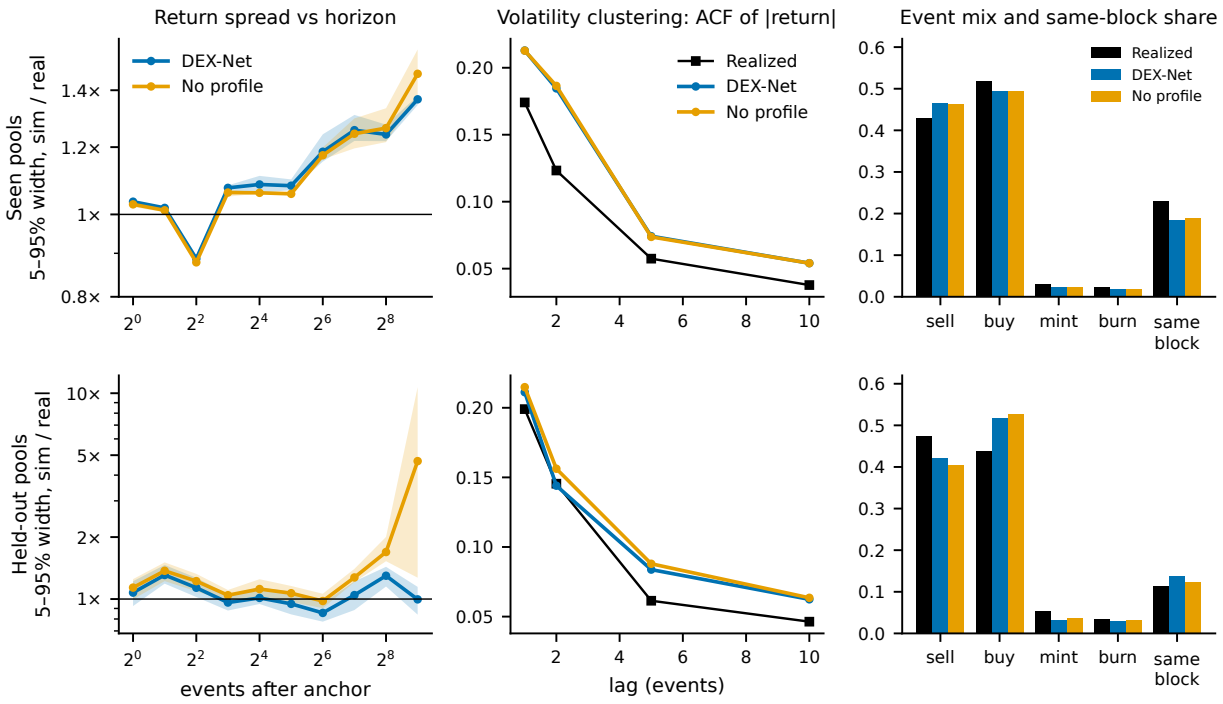


Figure 4: Stylized facts after test anchors, seen (top) and held-out (bottom) pools. Left: width of the 5–95% range of cumulative return, simulated over realized, by horizon (shaded: seed range). Center: autocorrelation of absolute per-event return. Right: event mix and same-block share.

at every horizon. Figure 4 computes this ratio per seed and averages seeds; for the no-profile model one seed's ratio is about 10× at 512 events, so its average (about 5×) is far above the pooled ratio of Table 5 (1.78×), which we treat as primary. Volatility clustering is too strong in every model: the autocorrelation of absolute per-event returns exceeds the realized value at every lag (seen pools, lag 1: 0.21 against 0.17). PIT histograms (Figure S1, Appendix A) show no gross miscalibration of the return forecast but visible deviations at individual horizons.

Predicted asymmetry is not directionally informative. For each forecast we take the 10th, 50th and 90th percentiles (q10, q50, q90) of the simulated return; its quantile skew is (q90 + q10 − 2 q50)/(q90 − q10), positive when the upside tail is longer. A breakout is a realized return above the forecast's q90 (upside) or below its q10 (downside), and an anchor's realized tail asymmetry is +1 for an upside breakout, −1 for a downside one and 0 otherwise. Skew and the q10/q90 thresholds are estimated from different paths, to avoid a spurious negative relation. The correlation between predicted skew and realized tail asymmetry lies between −0.12 and 0.00 at every horizon and pool group, with every confidence interval including zero. Restricting attention to the anchors with the most extreme predicted skew (top or bottom 20%) does not help. On held-out pools, where 52–63% of all breakouts are upward, anchors in the bottom 20% of predicted skew were followed by upside breakouts 65–86% of the time (19 anchors per group; small samples). Predicted skew follows the preceding trend (correlation 0.07–0.20 between the prior move and predicted skew), but realized outcomes after strong trends show no consistent direction, so following the trend does not improve the forecast.

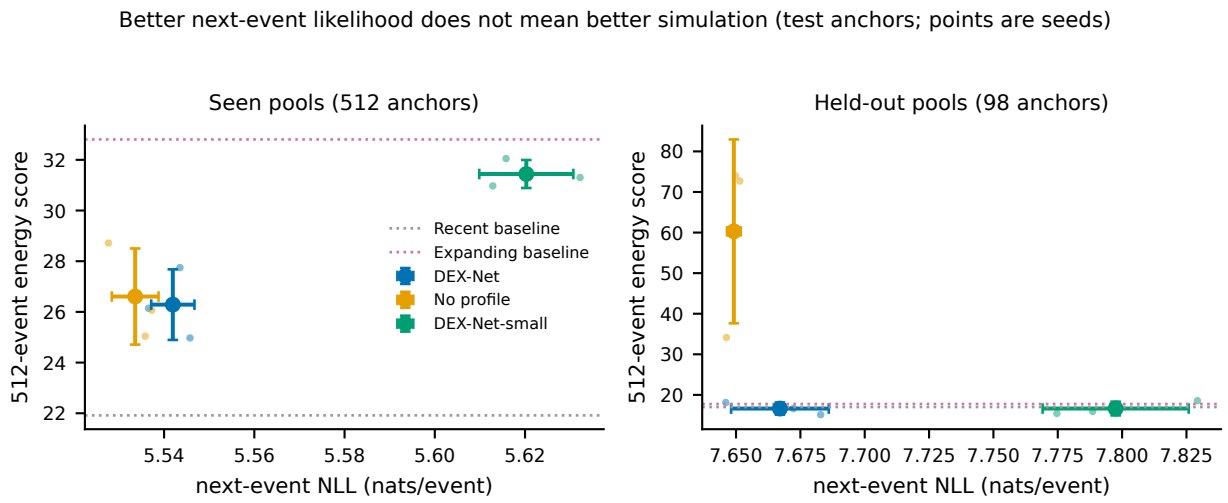


Figure 5: Next-event NLL against 512-event energy on test anchors for the three rollout-evaluated families; large points are seed means with standard deviations, small points are individual seeds; dotted lines are the history-based forecasters.

6.5 Likelihood is not simulation quality

Figure 5 places each model family by next-event NLL and 512-event energy. On held-out pools, likelihood does not predict rollout quality: the no-profile model has the lowest NLL but the worst rollouts, and DEX-Net-small has a higher NLL than the no-profile model but rollouts within 4% of DEX-Net's. The failures that matter for simulation are rare and state-dependent (thin liquidity, oversized swaps in a depleted book), so they carry little weight in an average one-step likelihood but compound over a rollout. Model selection for simulators should therefore use rollout scores, or at least check them. The general pattern is known (Section 2). Training conditions only on real histories, which keep the model on-distribution (exposure bias; Bengio et al., 2015; Ranzato et al., 2016), so the liquidity profile adds little to one-step likelihood; it matters once a rollout drifts into states that real histories rarely show. What is specific to AMMs is the feedback loop: liquidity thins, sampled swaps become oversized relative to it, and liquidity thins further. Neural point-process models are usually compared by next-event likelihood, and for simulators that comparison should be checked against rollouts.

6.6 LP-facing evaluation

To test the simulator on what an LP earns, we place a hypothetical marginal position over $\pm 1\%$ around the price at each anchor. For every simulated path, the realized path and each history window we compute the position's fee income, its divergence loss (value against holding its initial tokens) and their sum, its P&L against holding, in bp of the initial position value. A marginal position's fee per unit of liquidity depends only on the price path: a swap that moves $\sqrt{P}$ from a to b within the range pays $\frac{f}{1-f}\,|b-a|$ in quote when the price rises and $\frac{f}{1-f}\,|1/b-1/a|$ in base when it falls, where f is the fee tier; because $\sqrt{P}$ moves monotonically within a swap, each swap's start and end prices suffice. Results are gross of any protocol fee share. We use holding the initial tokens, not a rebalancing strategy, as the benchmark: LVR requires an external price, which the simulator does not produce. Forecasts are scored by CRPS (Table 7).

Table 7: LP-facing evaluation, ±1% position. CRPS (bp of position value, lower is better) of P&L versus holding; DEX-Net minus the recent baseline with 95% bootstrap CI; DEX-Net 90%-interval coverage of the realized P&L; fee-income CRPS for DEX-Net / recent baseline; real median fee income in bp. Gross of any protocol fee share.

Period	Pools	Events	Anchors	Real fee	DEX-Net	No profile	DEX-Net-small	Recent	Expanding	DEX-Net − recent	Coverage	Fee CRPS
Test	held out	32	98	3.0	2.50	3.12	2.63	**2.48**	2.81	+0.03 [-0.27, +0.29]	83%	2.1 / 2.3
Test	held out	128	98	15.1	12.11	22.99	12.23	**11.66**	12.17	+0.45 [-0.53, +1.52]	82%	5.6 / 5.8
Test	held out	512	98	37.5	38.61	117.50	41.41	**35.30**	36.27	+3.31 [+0.13, +7.85]	81%	18.2 / 18.2
Test	seen	32	512	1.9	**3.96**	3.99	4.20	4.12	4.52	-0.16 [-0.31, -0.02]	89%	1.5 / 1.7
Test	seen	128	512	8.3	14.88	**14.86**	16.63	15.09	15.67	-0.22 [-0.67, +0.21]	89%	4.8 / 4.9
Test	seen	512	512	24.4	41.70	40.39	53.05	**37.94**	39.54	+3.76 [+1.40, +6.87]	86%	14.7 / 14.3
Validation	held out	32	95	6.9	5.95	6.57	**5.73**	5.78	6.19	+0.17 [-0.66, +1.06]	84%	2.7 / 3.4
Validation	held out	128	95	17.0	17.37	23.20	16.71	17.12	**16.59**	+0.24 [-1.32, +1.91]	86%	8.4 / 8.8
Validation	held out	512	95	37.6	65.97	114.01	66.02	**63.76**	65.02	+2.21 [-1.51, +6.38]	69%	26.4 / 26.9
Validation	seen	32	512	3.2	8.78	**8.71**	8.99	9.21	9.42	-0.43 [-0.79, -0.14]	87%	1.8 / 2.3
Validation	seen	128	512	11.7	**20.95**	21.15	21.69	21.71	21.86	-0.76 [-1.44, -0.12]	86%	6.0 / 6.7
Validation	seen	512	512	29.4	52.01	54.43	57.84	51.62	**51.09**	+0.39 [-1.10, +1.85]	82%	18.5 / 19.3

DEX-Net's fee-income forecasts are better than the recent baseline's in 10 of 12 horizon, period and pool-group cells; the two are equal at the reported precision in one, and the recent baseline is slightly better in one (seen test pools at 512 events: CRPS 14.3 against 14.7). For P&L, DEX-Net is better at short horizons on seen pools (test, 32 events: by 0.16, CI 0.02 to 0.31; validation, 32 and 128 events). Elsewhere the two are within 4% of each other (with intervals as wide as −12% to +18% on held-out pools), except at 512 events in the test period, where the recent baseline is 9.0% better on seen pools (CI 3.4% to 16.5%) and 8.6% better on held-out pools (CI 0.3% to 20.3%). The difference comes from the divergence component, consistent with DEX-Net's too-wide long-horizon price spread on seen pools (Section 6.4), and DEX-Net's 90% P&L intervals cover only 69–81% of held-out outcomes at 512 events. On held-out pools the recent baseline is much better than the no-profile model (P&L CRPS 35.3 against 117.5 at 512 events in the test period). A narrower ±0.2% position gives qualitatively the same results.

With the path reweighting of Section 6.2, which was fitted on validation endpoints only and is therefore out of sample here, DEX-Net's P&L forecasts are better than the recent baseline's on seen test pools at every horizon (by 0.23 at 32 events, CI 0.08 to 0.38; 0.58 at 128 events, CI 0.14 to 1.03; 5.15 at 512 events, CI 0.52 to 11.66), and on held-out pools at 512 events the two are level (difference 0.51 in DEX-Net's favor, CI −4.36 to +5.18). The gain comes from the divergence component (seen pools, 512 events: CRPS 24.5 against 29.0 for the recent baseline), that is, from correcting the too-wide long-horizon price spread. Held-out 90% P&L intervals remain too narrow (81–83% coverage).

6.7 Ablations: exact mechanics and joint training

Exact mechanics. To measure what the replica contributes, we train the no-profile model with learned transitions instead, with three seeds and the same schedule and selection. After each sampled event, two added heads give its log price return (swaps only: a zero indicator, then a spline flow on log|return|, whose sign follows the side) and its change in log active liquidity (a zero indicator, then a spline flow). There is no tick book or position inventory: buys are not truncated at the pool's capacity, burn ranges are not restricted to existing positions, and burn size is drawn relative to active liquidity, like mint size. Rollouts track the price, active liquidity and virtual reserve from the learned changes and are scored on the same anchors and endpoints (Table 8). At most 0.15% of its paths fail within 512 events.

On seen pools both exact models are better than the learned-transition model at every horizon,

Table 8: Mechanics ablation. Rollout energy (lower is better) of DEX-Net, the no-profile model, and the no-profile model with learned transitions in place of the exact replica, on the Table 4 anchors (3 seeds × 32 paths); paired differences with 95% bootstrap CI over anchors.

Period	Pools	Events	DEX-Net	No profile	Learned transitions	Learned − no profile	Learned − DEX-Net
Test	seen	32	3.00	3.00	3.11	+0.11 [+0.03, +0.20]	+0.11 [+0.02, +0.20]
Test	seen	128	7.88	7.84	8.04	+0.20 [+0.04, +0.36]	+0.16 [−0.02, +0.34]
Test	seen	512	26.29	26.61	33.16	+6.55 [+5.71, +7.40]	+6.87 [+5.95, +7.78]
Test	held out	32	2.75	3.43	2.84	−0.59 [−1.46, +0.02]	+0.08 [−0.12, +0.35]
Test	held out	128	6.38	12.46	6.39	−6.07 [−14.15, −1.25]	+0.01 [−0.52, +0.49]
Test	held out	512	16.63	60.30	18.48	−41.82 [−76.25, −19.88]	+1.84 [+0.18, +3.34]
Validation	seen	32	5.30	5.33	5.44	+0.11 [−0.01, +0.25]	+0.14 [+0.03, +0.26]
Validation	seen	128	12.11	12.23	12.76	+0.53 [+0.11, +1.03]	+0.65 [+0.20, +1.21]
Validation	seen	512	37.94	39.27	43.96	+4.68 [+3.12, +6.20]	+6.02 [+4.37, +7.92]
Validation	held out	32	6.64	7.24	6.49	−0.75 [−2.72, +0.43]	−0.15 [−1.10, +0.49]
Validation	held out	128	11.81	15.71	10.98	−4.73 [−10.94, −1.03]	−0.83 [−3.07, +0.60]
Validation	held out	512	33.03	80.30	28.67	−51.63 [−76.70, −29.76]	−4.36 [−9.68, −0.41]

significantly so at 512 events: the no-profile model is 19.8% better in the test period (26.61 against 33.16; CI 17.2% to 22.3%) and 10.6% better in validation (39.27 against 43.96; CI 7.1% to 14.1%). The exact model is better on seven of the eight seen pools in both periods. The difference is in liquidity and traded volume (test, scaled CRPS 1.1 against 2.7 for liquidity and 21.4 against 27.2 for volume), not in price return (11.0 for both). Without exact bookkeeping, the learned-transition model's liquidity drifts: 16.8% of its seen-pool paths lose more than 63% of active liquidity within 512 events, against 3.9% for the no-profile model and 1.6% of real outcomes.

On held-out pools the learned-transition model is better than the no-profile model, and neither it nor DEX-Net is better in both periods. It avoids the no-profile model's collapse on the stablecoin pool (512-event energy 5.5 against 127.5 in the test period; 4.3 against 168.5 in validation) and is better than DEX-Net there too (5.5 against 7.6; 4.3 against 17.3). On WBTC/WETH 0.3% DEX-Net is 15.5% better in the test period (21.4 against 25.4; CI 8.0% to 22.7%), and the two are level in validation (40.5 for the learned-transition model against 40.6; difference 0.4%, CI −7.6% to +8.9%). Over both held-out pools DEX-Net is 10.0% better in the test period (16.63 against 18.48; CI 1.0% to 18.1%), and the learned-transition model is 13.2% better in validation (28.67 against 33.03; CI 1.2% to 29.3%). The learned-transition model is also better than DEX-Net on the seen stablecoin pool (USDC/USDT 0.01%: 3.6 against 5.4 in test; 2.1 against 2.5 in validation). We attribute this to how concentrated stablecoin liquidity is: with most of it in one or two ticks, a slightly oversized sampled swap carries the replica's price past the edge of that range into thin liquidity, while the learned model predicts the tiny price change directly. The liquidity profile reduces this amplification but does not remove it.

In summary, exact mechanics make the simulated pool more accurate on the seen pools and keep its liquidity realistic. They also amplify errors in sampled swap sizes, and the liquidity profile reduces that amplification. On an unfamiliar pool the learned-transition model is better than the exact model without the profile.

Joint training. All other models are trained jointly on the eight seen pools. To test whether this helps an individual pool, we train DEX-Net on one pool with the same budget (202 million examples in 97 rounds: 733 passes over DAI/WETH 0.3% and 24 over USDC/WETH 0.05%), selecting checkpoints by that pool's validation NLL, with three seeds each (Table 9). On DAI/WETH 0.3%, the smallest seen pool (275,445 training events), the joint model has a lower next-event NLL than

Table 9: Joint-training ablation. DEX-Net trained jointly on the eight seen pools against DEX-Net trained on one pool with the same budget, evaluated on that pool: test-period next-event NLL (nats/event, 3-seed mean; difference with s.e.) and rollout energy on the pool's 64 test and 64 validation anchors, with paired 95% bootstrap CI.

Pool	Measure	Joint	Single pool	Single − joint
DAI/WETH 0.3%	NLL, test	5.606	5.676	+0.071 (s.e. 0.008)
DAI/WETH 0.3%	Energy, test, 32 events	6.25	6.22	−0.03 [−0.37, +0.24]
DAI/WETH 0.3%	Energy, test, 128 events	18.05	19.46	+1.41 [−0.01, +2.43]
DAI/WETH 0.3%	Energy, test, 512 events	62.23	76.57	+14.35 [+9.00, +18.54]
DAI/WETH 0.3%	Energy, validation, 512 events	90.47	104.15	+13.68 [+7.51, +21.29]
USDC/WETH 0.05%	NLL, test	4.792	4.813	+0.021 (s.e. 0.003)
USDC/WETH 0.05%	Energy, test, 32 events	1.69	1.67	−0.02 [−0.06, +0.02]
USDC/WETH 0.05%	Energy, test, 128 events	4.33	4.23	−0.11 [−0.26, +0.05]
USDC/WETH 0.05%	Energy, test, 512 events	16.27	16.25	−0.02 [−0.78, +0.78]
USDC/WETH 0.05%	Energy, validation, 512 events	15.09	16.01	+0.92 [−0.00, +1.84]

the single-pool model, by 0.071 nats per event (s.e. 0.008) and in every component of it. It is also better at 512 events, by 18.7% in the test period (62.23 against 76.57; CI 11.8% to 24.2%) and 13.1% in validation (90.47 against 104.15; CI 7.2% to 20.4%), and by 7.2% at 128 events in test (CI −0.1% to +12.5%); at 32 events the two are level (the single-pool model is 0.5% better, CI −3.8% to +5.9%). The single-pool model's validation NLL is lowest at rounds 8–11 of 97, so it begins to overfit early. On USDC/WETH 0.05%, the largest pool (8.4 million training events), the joint model still has a lower NLL (by 0.021, s.e. 0.003) and is 2.8% better at 512 events over both periods (15.68 against 16.13; CI −1.0% to +6.6% over 128 anchors). By period, the single-pool model is 0.1% better in test and the joint model is 5.7% better in validation; this sample cannot distinguish no difference from a difference of about 7%. By endpoint, the single-pool model is better on liquidity (CRPS 0.095 against 0.114 in log liquidity) and the joint model on traded volume (0.0130 against 0.0136 in log volume), with return and timing about equal. The single-pool model's checkpoints are selected at rounds 84–97. Joint training therefore gives each tested pool a lower NLL and gives a pool with a short history better long rollouts; for a pool with millions of events of its own, the rollout difference is small and mixed across endpoints. Fine-tuning the joint model on a single pool (starting from the joint checkpoint, 10 rounds at a learning rate of 2×10^{-5}, selected by that pool's validation NLL) does not reliably help. We fine-tune on each of the seven non-stablecoin seen pools, from 275,445 to 8.4 million training events, with three seeds each (Table 10). At 512 events in the test period, the fine-tuned model is significantly better than the joint model on three pools (DAI/WETH 0.3%, DAI/WETH 0.05% and USDC/WETH 0.05%, by 5.6–10.0%), and the joint model is significantly better on two (WBTC/WETH 0.05% and USDT/WETH 0.3%, by 16.5% and 17.1%). In validation the joint model is significantly better on two pools and the fine-tuned model on none. Averaged over the seven pools, the joint model is about 2% better in test and 3% better in validation, and it has the lower test NLL on five pools. Only on the largest pool, USDC/WETH 0.05%, is the fine-tuned model better in both periods, and the gain from fine-tuning shows no trend with pool size over this range. On several pools validation NLL is lowest within the first few fine-tuning rounds, which suggests that training further on one pool's training period fits that period at the expense of later ones, while training across eight pools carries over between periods better. For a pool with its own history we therefore recommend the jointly trained model, which is also the model for a pool without history.

Table 10: Fine-tuning the joint model on one pool. For each non-stablecoin seen pool: training events; how much lower the fine-tuned model's test-period NLL is than the joint model's (nats/event; negative: the joint model's is lower; s.e. in parentheses); and how much better the fine-tuned model's 512-event energy is than the joint model's, relative to the joint model (negative: the joint model is better), on the pool's 64 test and 64 validation anchors, with paired 95% bootstrap CI. Fine-tuning: 10 rounds at learning rate 2×10^{-5} from the joint checkpoint, best round by that pool's validation NLL, three seeds.

Pool	Training events	Fine-tuned NLL lower by	512 events, test	512 events, validation
DAI/WETH 0.3%	275,445	−0.000 (0.004)	+10.0% [+6.6%, +14.2%]	−0.0% [−5.9%, +5.2%]
USDC/WETH 0.3%	795,457	−0.018 (0.008)	−1.5% [−5.0%, +2.1%]	−1.6% [−3.8%, +0.5%]
USDT/WETH 0.3%	796,809	−0.050 (0.035)	−17.1% [−22.8%, −11.2%]	−3.9% [−8.7%, +0.9%]
DAI/WETH 0.05%	1,160,931	−0.003 (0.002)	+5.6% [+1.6%, +9.5%]	−1.9% [−6.2%, +1.6%]
WBTC/WETH 0.05%	1,298,702	−0.019 (0.015)	−16.5% [−22.7%, −10.6%]	−7.3% [−13.7%, −1.3%]
USDT/WETH 0.05%	4,746,323	−0.010 (0.003)	+1.1% [−2.9%, +5.6%]	−6.7% [−12.6%, −2.0%]
USDC/WETH 0.05%	8,416,528	+0.012 (0.004)	+5.6% [+2.1%, +10.3%]	+3.0% [−1.1%, +7.5%]

7 Limitations

- **Own-pool context only.** The model sees one pool's history; cross-pool and centralized-exchange information, which drives arbitrage flow, is absent.
- **Simulating the pool, not interventions.** DEX-Net simulates the pool's order flow; it does not model how that flow would respond to a new participant. The LP evaluation does not add the hypothetical position to the simulated pool, so its results apply only to positions too small to change the flow, and using the simulator to test strategies large enough to move the pool would be an untested counterfactual.
- **Scope.** All pools are high-volume Ethereum mainnet pairs (eight for training, two held out); long-tail pools, layer-2 deployments and Uniswap v4 are untested. Held-out evidence rests on two pools, and the liquidity profile's largest effect is on one of them; the comparison with single-pool training covers two of the eight seen pools, and the fine-tuning comparison seven.
- **Remaining failures.** Skew is not directionally informative; swap-size tails are still heavier than real; 15% of held-out paths still collapse against 5% of real outcomes (42% against 15% on the stablecoin pool), and on stablecoin pools a model with learned transitions is more accurate, so exact mechanics still amplify swap-size errors there; volatility clustering is too strong; JIT is too rare, with deposits too small for the swaps they accompany, and too sticky, mostly because the model burns a different position than the one just minted; without reweighting, a recent-history baseline forecasts long-horizon LP P&L slightly better, and 90% P&L intervals are too narrow on held-out pools even with it.
- **Baselines.** On the seen-pool test period, the recent-history forecaster is better at 512 events; on held-out pools the difference is within 5% but the intervals span roughly −15% to +23%, so we cannot say which is better. We do not evaluate price-only or agent-based simulators directly.

8 Conclusion

We asked three questions. First, computing the pool mechanics exactly, instead of learning them, makes the simulation more accurate on the eight training pools: exact models forecast 512 events

ahead 11–20% better than a model that learns price and liquidity changes, and their liquidity stays realistic. The exception is stablecoin pools, where the learned model is more accurate. Second, to simulate a pool it was not trained on, the model needs the current liquidity profile as an input: active liquidity at 17 price offsets within ±4% of the current price, read from the reconciled ledger in training and recomputed from the simulated book after every sampled event. Without it, small errors in swap size compound through the exact mechanics. On a held-out stablecoin pool, 68% of 512-event paths lose most of their liquidity without the profile and 42% with it (real: 15%), and the energy score is 127 without it and 7.6 with it. Across both held-out pools the profile reduces price moves above 100 bp from 0.97% to 0.20% of swaps (real: 0.03%) and narrows the 512-event price spread from about 1.8 times realized to 1.01 times in the test period and 0.80 times in validation. Third, for a pool with its own history, the best choice is the model trained jointly on all pools. It gives both tested pools a lower NLL than training on the pool alone, and on the smallest pool 13–19% better long rollouts. Fine-tuning it on one pool was not reliably better: across seven pools the joint model was about 2% better on average at 512 events in the test period and 3% in validation, and had the lower NLL on five, so training further on one pool's own history appears to fit that period at the expense of later ones. On method, next-event likelihood is a poor guide to simulation quality: the profile changes it by only 0.018 nats on held-out pools while rollouts change a great deal, and the model with the lowest NLL has the worst rollouts, so rollout scores, not likelihood, should guide model selection for simulators. The remaining failures (heavy swap-size tails, directionless skew, too-rare and too-sticky JIT, too-narrow long-horizon P&L intervals, residual error amplification on stablecoin pools) point to three next steps: training on the model's own multi-step rollouts so that compounding errors are penalized directly (an earlier attempt, before the liquidity profile, gave little benefit), selecting checkpoints by rollout scores rather than likelihood, and post-hoc calibration of simulated paths, such as the path reweighting and forecast blend of Section 6.2.

Code and data. Code and model weights (Apache-2.0) and the reconciled dataset (CC BY 4.0) are available at https://github.com/Claybarn/DEX-Net.

A Supplementary results

Path reweighting. Each path i in a seed's ensemble of n paths receives weight

$$w_i = 0.9\,\mathrm{softmax}_i(s_i) + 0.1/n \quad \text{with score} \quad s_i = 2\tanh\!\big(c^\top x_i/\sqrt{k}\big).$$

The $k = 25$ features x_i are the signed and absolute deviations of each of the four endpoints at each of the three horizons from the ensemble median, scaled by the typical ensemble spread on the fitting data, plus a flag for a failed path. The floor and the bounded score keep every path in the forecast and limit any path's weight. The global model fits one coefficient vector c; the conditional model adds a small network of anchor context. We minimize the mean over horizons of the weighted energy score divided by its uniform-weight value, plus λ times the KL divergence from uniform weights (Adam, learning rate 0.02, 400 steps, three fitting seeds averaged). The model is fitted on the earliest 75% (by time) of seen-pool validation anchors, $\lambda \in \{0.001, 0.01, 0.1\}$ is chosen on the latest 25% ($\lambda = 0.1$ for the global model), and held-out pools are never used. On test anchors, weighted ensembles are scored with the weighted analogue of the fair estimator:

$$\mathbb{E}_w|X - y| - \frac{\sum_{i\neq j} w_i w_j |X_i - X_j|}{2\,(1 - \sum_i w_i^2)}$$

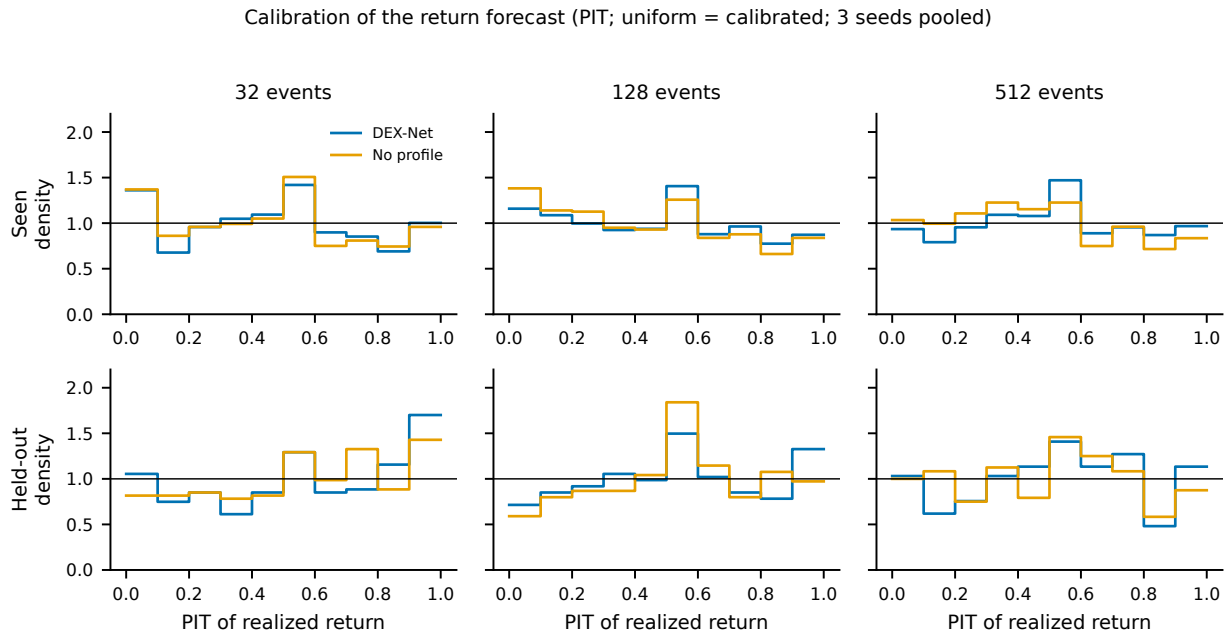


Figure S1: Probability integral transform (PIT) of the realized return under each model's forecast, pooled over three seeds; a calibrated forecast gives a uniform histogram (density 1).

, which equals the usual fair estimator for uniform weights. The fair correction assumes exchangeable ensemble members; here the weights are learned functions of the paths and of ensemble statistics, so we do not claim the weighted version is unbiased. We use it so that reweighted and unweighted ensembles are scored on the same footing, with identical results when the weights are uniform.

Implementation note. Interval endpoints of the discretized range distributions are computed in double rather than single precision: in single precision, the two edges of a one-tick interval far from the current tick can round to the same value.

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